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RATIO AND PROPORTION

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Introduction

Ratios are the comparisons made between two set of numbers

Example :
There are 80 female employees and 100 male employees in a firm, then the ratio of male to female employees is 10 : 8 → 5:4

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Concept and Definitions

The compounded ratio of the two ratios a : b and c : d is the ratio ac : bd, and that of a : b, c : d and e : f is the ratio ace : bdf.

For ratios m : n and p : q; the compound ratio is (m × p) : (n × q).

For ratio m : n, p : q and r : s; the compound ratio is (m × p × r) : (n × q × s).

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Key Formulae

The duplicate ratio of the ratio m : n =
Compound ratio of
m : n and m : n
= (m × m) : (n × n)
= m^2: n^2
The sub duplicate ratio m : n is the ratio $\sqrt{m} : \sqrt{n}$. So, the sub duplicate ratio of the ratio m2 : n2 is the ratio m : n.

Key Formulae

The triplicate ratio of the ratio m : n

= Compound ratio of m : n, m : n and m : n

= (m × m × m) : (n × n ×n)

= m3 : n3.

The sub triplicate ratio m : n is the ratio $\sqrt[3]{m} : \sqrt[3]{n}$.

PROPORTIONS :

If a : b = c : d, we write a : b :: c : d and we say a , b, c ,d are in proportion

Here a and d are called extremes and b and c are called mean terms

a x d = b x c

Key Formulae

COMPOUND RATIO :

If two ratios A : B = 3 : 4 and B : C = 4 : 5 are given and we have to find the A: B: C

Now, We have to follow the step

A * B : B * B : B *C
3 *4 : 4 *4 : 4 *5
12 : 16 : 20

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Example

If a : b = 5 : 9 and b : c = 4 : 7, find a : b : c

a : b : c = 5 * 4 : 9 * 4 : 9 * 7

= 20 : 36 : 63

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Questions

Ques.1 The perimeter of a rectangle is 64 cm. If the ratio of the lengths of two adjacent sides is 7: 9, find the lengths of these sides.

a) 24 cm, 28 cm

b) 14 cm, 18 cm

c) 7 cm, 9 cm

d) None of these

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Questions

Ques.2 The ratio of the length and the breadth of a rectangle is 3 : 5 and its area is 1.35 cm2. Find the length of the rectangle.

a) 9 cm

b) 0.9 cm

c) 0.09 cm

d) None of these

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Questions

Ques.3 The ratio of the present ages of John and Jim is 5 : 3. Four years hence it will be 3 : 2. Find the present age of John.

a) 5 years

b) 7 years

c) 10 years

d) None of these

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Questions

Ques.4 There are 145 students in the first three standards. The ratio of number of students in the first and the second standards is 2 : 3, while that of students in standards second and third is 4: 3. Find the number of students in 2nd standard.

a) 40

b) 60

c) 45

d) None of these

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Questions

Ques.5 The ratio of the measures of ∠A and ∠B of a triangle is 3 : 2.The ratio of measures of ∠B and ∠C is 4 : 5. Find the measure of largest angle of the triangle ABC.

a) 72°

b) 78°

c) 48°

d) 60°

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Questions

Ques.6 A piece of string 70 cm in length was cut into pieces, the ratio of whose lengths was 3: 7. Find the length of longest piece

a) 21 cm

b) 70 cm

c) 49 cm

d) 7 cm

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Questions

Ques.7 Divide Rs. 672 in the ratio 5 : 3.

a) 420,252

b) 421,251

c) 500,172

d) None of these

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Questions

Ques.8 A bag contains 50p,25p and 10p coins in the ratio 5 : 9 : 4,amounting to Rs.206.Find the number of coins each type

a) 200,360 and 160

b) 210,350 and 160

c) 200,300,120

d) None of these

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Questions

Ques.9 If Rs.782 be divided into three parts , proportional to 1/2 :2/3 :3/4 then the first part is

a) 204

b) 402

c) 212

d) None of these

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Questions

Ques.10 If sum of 4 times a number A and 3 times a number B is equals to sum of number B and 7 times of the number A, then what is the value of A:B ?

a) 2: 3

b) 3 : 2

c) 4 : 5

d) None of these

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Questions

Ques11. Incomes of A and B are in the ratio of 2:3 and each of them saves Rs.300 . Find A's income, If the ratio of their expenditure is 1:2?

a) 300

b) 600

c) 750

d) None of these

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Questions

Ques12. A sum of money is to be distributed among A, B, C, D in the proportion of 5 : 2 : 4 : 3. If C gets Rs. 1000 more than D, what is B's share?

a) 300

b) 600

c) 2000

d) None of these

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Questions

Ques13. Seats for Mathematics, Physics and Biology in a school are in the ratio 5 : 7 : 8. There is a proposal to increase these seats by 40%, 50% and 75% respectively. What will be the ratio of increased seats?

a) 2:3:4

b) 6:7:8

c) 6:8:9

d) None of these

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Questions

Ques14. In a mixture 60 liters, the ratio of milk and water 2 : 1. If this ratio is to be 1 : 2, then the quantity of water to be further added is:

a) 30 L

b) 60 L

c) 20 L

d) None of these

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Questions

Ques15. If 6m – n = 4m + 13n, find the value of 2m + n : 2m – 3n.

a) 14:11

b) 15:14

c) 15:11

d) None of these

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Thank You